

MODEL QUESTION PAPER

TED (10)4083

Register Number.....

(REVISION-2010)

Signature.....

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

(Common to EI & IT)

BIOMEDICAL INSTRUMENTS

Time: 3Hours

(Maximum marks: 100)

Part-A

I (Answer all questions in one or two sentence. Each question carries 2 marks)

- 1) What is Sodium Pump?
- 2) Write the significance of QRS Complex in an ECG wave form?
- 3) What is electro myogram?
- 4) State the use of respirator
- 5) State the need of Biotelemetry

(5 x 2 =10)

Part-B

II (Answer any 5 questions. Each questions carries 6 marks)

- 1) Explain generation of resting and action potentials in a cell.
- 2) Explain the working of a blood flow meter in which the electrical conductivity property of blood is utilized
- 3) Write the different wave frequencies in EEG and draw the block diagram of an EEG Machine.
- 4) What is a pace maker? Draw the block Diagram of Ventricular synchronized demand pace maker
- 5) Explain the working of short wave diathermy Machine.
- 6) Illustrate the generation of X rays.
- 7) Distinguish A-Scan and B-Scan Ultrasonic Imaging system.

(5 x 6 =30)

Part-C

(Answer one full question from each module. Each question carries 15 marks)

Module-I

- III (a) Explain the working principle of optical fiber temperature sensor (7)
(b) Describe blood pressure measurement using sphygmomanometer (8)

OR

- IV (a) Explain the working principle of strain gauge type respiration sensor (7)
(b) Describe the working principle of photo electric Pulse transducer (8)

MODULE-II

- V (a) Describe the 10-20 EEG electrode configuration. (7)
(b) Describe the working of EMG machine with block diagram (8)

OR

- VI (a) Draw the different Lead configurations used in ECG measurement (7)
(b) Draw the block diagram and explain the working of EEG machine (8)

MODULE- III

- VII (a) With a detailed schematic diagram explain the working of DC Defibrillator (7)
(b) Describe the working of electrical conductivity blood cell counter. (8)

OR

- VIII (a) Explain the working of ultrasonic diathermy unit (7)
(b) Describe the functions of hemodialysis machine (8)

MODULE- IV

- IX (a) Explain the working principle of CAT scanner with figure (10)
(b) Distinguish Macro shock and Micro shock (5)

OR

- X (a) Summarize the effect of electrical current on human body (7)

(b) Explain any two application of biotelemetry

(8)